

Complete Nextcloud 34 Installation Guide on Ubuntu Server 24.04

Environment

Item	Value
Server OS	Ubuntu Server 24.04
Source	\\tns\Infrastructure\Network Configurations and Plans\Subsidiaries\KITCO\LAB\Next Cloud
Server IP	172.16.0.237(U:kit P: ASDqwe@000237)
Public FQDN	kitcld.kayson-ir.com
Internal AD Domain	kayson-ir.com
Web Server	Apache
Database	MariaDB
SSL	Let's Encrypt

Phase 1 — DNS Configuration

External DNS

Create an A record:

```
Host: kitchd
Type: A
Value: YOUR_PUBLIC_IP
```

Purpose:

Internet users can reach your Nextcloud server.

Verify:

```
nslookup kitchd.kayson-ir.com
```

Internal AD DNS

On Domain Controller:

DNS Manager

Create:

```
Forward Lookup Zones
└─ kayson-ir.com
   └─ New Host (A)
      Name: kitchd
      IP: 172.16.0.237
```

Purpose:

LAN clients connect directly without hairpin NAT.

Verify:

```
nslookup kitchd.kayson-ir.com
```

Expected:

```
172.16.0.237
```

Phase 2 — Firewall

Forward:

```
TCP 80 → 172.16.0.237
TCP 443 → 172.16.0.237
```

Purpose:

HTTP and HTTPS access.

Phase 3 — Update Ubuntu

Update package lists:

```
sudo apt update
```

Explanation:

Part	Meaning
sudo	Run as administrator
apt	Ubuntu package manager
update	Download latest package indexes

Install updates:

```
sudo apt upgrade -y
```

Explanation:

Part	Meaning
upgrade	Install available updates
-y	Automatically answer yes

Phase 4 — Install Apache

```
sudo apt install apache2 -y
```

Verify:

```
systemctl status apache2
```

Explanation:

Part	Meaning
systemctl	Service manager
status	Show service state
apache2	Service name

Expected:

```
active (running)
```

Phase 5 — Install MariaDB

```
sudo apt install mariadb-server -y
```

Verify:

```
systemctl status mariadb
```

Expected:

```
active (running)
```

Phase 6 — Secure MariaDB

Run:

```
sudo mysql_secure_installation
```

Purpose:

Remove insecure defaults.

Recommended:

```
Set root password → Y
Remove anonymous users → Y
Disallow remote root → Y
Remove test database → Y
Reload privilege tables → Y
```

Phase 7 — Create Nextcloud Database

Open MariaDB:

```
sudo mysql
```

Explanation:

Login to MariaDB as root.

Create database:

```
CREATE DATABASE nextcloud
CHARACTER SET utf8mb4
COLLATE utf8mb4_general_ci;
```

Purpose:

Create storage for Nextcloud data.

Create user:

```
CREATE USER 'nextclouduser'@'localhost'
IDENTIFIED BY 'StrongPassword123!';
```

Purpose:

Create dedicated database account.

Grant permissions:

```
GRANT ALL PRIVILEGES ON nextcloud.* TO 'nextclouduser'@'localhost';
```

Purpose:

Allow Nextcloud full access to its database.

Apply:

```
FLUSH PRIVILEGES;
```

Exit:

```
EXIT;
```

Phase 8 — Install PHP

```
sudo apt install -y \  
php \  
libapache2-mod-php \  
php-mysql \  
php-gd \  
php-curl \  
php-mbstring \  
php-intl \  
php-xml \  
php-zip \  
php-bcmath \  
php-gmp \  
php-imagick \  
php-apcu \  
php-redis
```

Purpose of important modules:

Module	Purpose
php-mysql	MariaDB connectivity
php-gd	Image processing
php-intl	Localization
php-zip	Archive support
php-imagick	Thumbnail generation
php-apcu	Cache
php-redis	Redis cache

Verify:

```
php -m
```

Shows installed PHP modules.

Phase 9 — Tune PHP

Edit:

```
sudo nano /etc/php/8.3/apache2/php.ini
```

Set:

```
memory_limit = 512M  
upload_max_filesize = 10G  
post_max_size = 10G  
max_execution_time = 360
```

Meaning:

Setting	Purpose
memory_limit	RAM per PHP process
upload_max_filesize	Largest upload

Setting	Purpose
post_max_size	Max HTTP upload size
max_execution_time	Script timeout

Restart Apache:

```
sudo systemctl restart apache2
```

Phase 10 — Download Nextcloud

```
cd /tmp
```

Meaning:

Change directory to temporary folder.

Download:

```
wget https://download.nextcloud.com/server/releases/latest.zip
```

Meaning:

Download latest Nextcloud release.

Extract:

```
unzip latest.zip
```

Move:

```
sudo mv nextcloud /var/www/
```

Phase 11 — Permissions

```
sudo chown -R www-data:www-data /var/www/nextcloud
```

Meaning:

Part	Meaning
chown	Change owner
-R	Recursive
www-data	Apache user

Phase 12 — Create Data Folder

Recommended:

```
sudo mkdir -p /data/nextcloud
```

Meaning:

Store files outside webroot.

Permissions:

```
sudo chown -R www-data:www-data /data/nextcloud
```

Phase 13 — Configure Apache

Create:

```
sudo nano /etc/apache2/sites-available/nextcloud.conf
```

Paste:

```
<VirtualHost *:80>
    ServerName kitcld.kayson-ir.com
    DocumentRoot /var/www/nextcloud

    <Directory /var/www/nextcloud>
        Require all granted
        AllowOverride All
        Options FollowSymLinks MultiViews
    </Directory>
</VirtualHost>
```

Enable:

```
sudo a2enmod rewrite headers env dir mime
```

Meaning:

Enable required Apache modules.

Enable site:

```
sudo a2ensite nextcloud.conf
```

Disable default:

```
sudo a2dissite 000-default.conf
```

Restart:

```
sudo systemctl restart apache2
```

Phase 14 — SSL

Install:

```
sudo apt install certbot python3-certbot-apache -y
```

Request certificate:

```
sudo certbot --apache -d kitcld.kayson-ir.com
```

Meaning:

Part	Meaning
certbot	Let's Encrypt client
--apache	Auto-configure Apache
-d	Domain name

Choose:

Redirect HTTP → HTTPS

Phase 15 — First Login

Open:

```
https://kitcld.kayson-ir.com
```

Fill:

```
Admin User: admin
Admin Password: *****
```

Database:

```
User: nextclouduser
Password: StrongPassword123!
Database: nextcloud
Host: localhost
```

Data folder:

```
/data/nextcloud
```

Click:

```
Install
```

Phase 16 — Verify Installation

Check status:

```
sudo -u www-data php /var/www/nextcloud/occ status
```

Meaning:

Part	Meaning
sudo -u www-data	Run as Apache user
php	Execute PHP
occ	Nextcloud admin console
status	Show installation status

Expected:

```
installed: true
```

Troubleshooting

Database Login Error

Test:

```
mysql -u nextclouduser -p nextcloud
```

If denied:

```
DROP USER IF EXISTS 'nextclouduser'@'localhost';

CREATE USER 'nextclouduser'@'localhost'
IDENTIFIED BY 'NewPassword';

GRANT ALL PRIVILEGES ON nextcloud.* TO 'nextclouduser'@'localhost';

FLUSH PRIVILEGES;
```

500 Internal Server Error

Check Apache:

```
sudo tail -100 /var/log/apache2/error.log
```

Check Nextcloud:

```
sudo tail -100 /data/nextcloud/nextcloud.log
```

PHP Memory Error

Check:

```
php -i | grep memory_limit
```

Increase:

```
memory_limit = 512M
```

Restart:

```
sudo systemctl restart apache2
```

Check Installed Apps

```
sudo -u www-data php /var/www/nextcloud/occ app:list
```

Lists enabled and disabled apps.

Repair Nextcloud

```
sudo -u www-data php /var/www/nextcloud/occ maintenance:repair
```

Purpose:

Fix database and application inconsistencies.

Common Useful Commands

Show Apache status:

```
systemctl status apache2
```

Show MariaDB status:

```
systemctl status mariadb
```

Restart Apache:

```
sudo systemctl restart apache2
```

Restart MariaDB:

```
sudo systemctl restart mariadb
```

Check disk space:

```
df -h
```

Check memory:

```
free -h
```

Check listening ports:

```
sudo ss -tulpn
```

This gives you a complete install, maintenance, and troubleshooting workflow from a fresh Ubuntu 24.04 server to a production-ready Nextcloud deployment.